

Tableau Training – Intermediate

Module 2: Calculations, Interactivity, and Advanced Charts

Welcome to Module 2!

What You Will Learn

This module bridges the gap from building basic charts to creating sophisticated, interactive analytical tools. We will focus on the "how" and "why" behind powerful Tableau features.

Key Objectives:

- Master data connections with Joins & Blends.
- Write powerful Level of Detail (LOD) Expressions.
- Create dynamic reports with Parameters & Sets.
- Build advanced charts (Maps, Dual-Axis, Treemaps).



Advanced Data Connections

Going beyond a single data file.

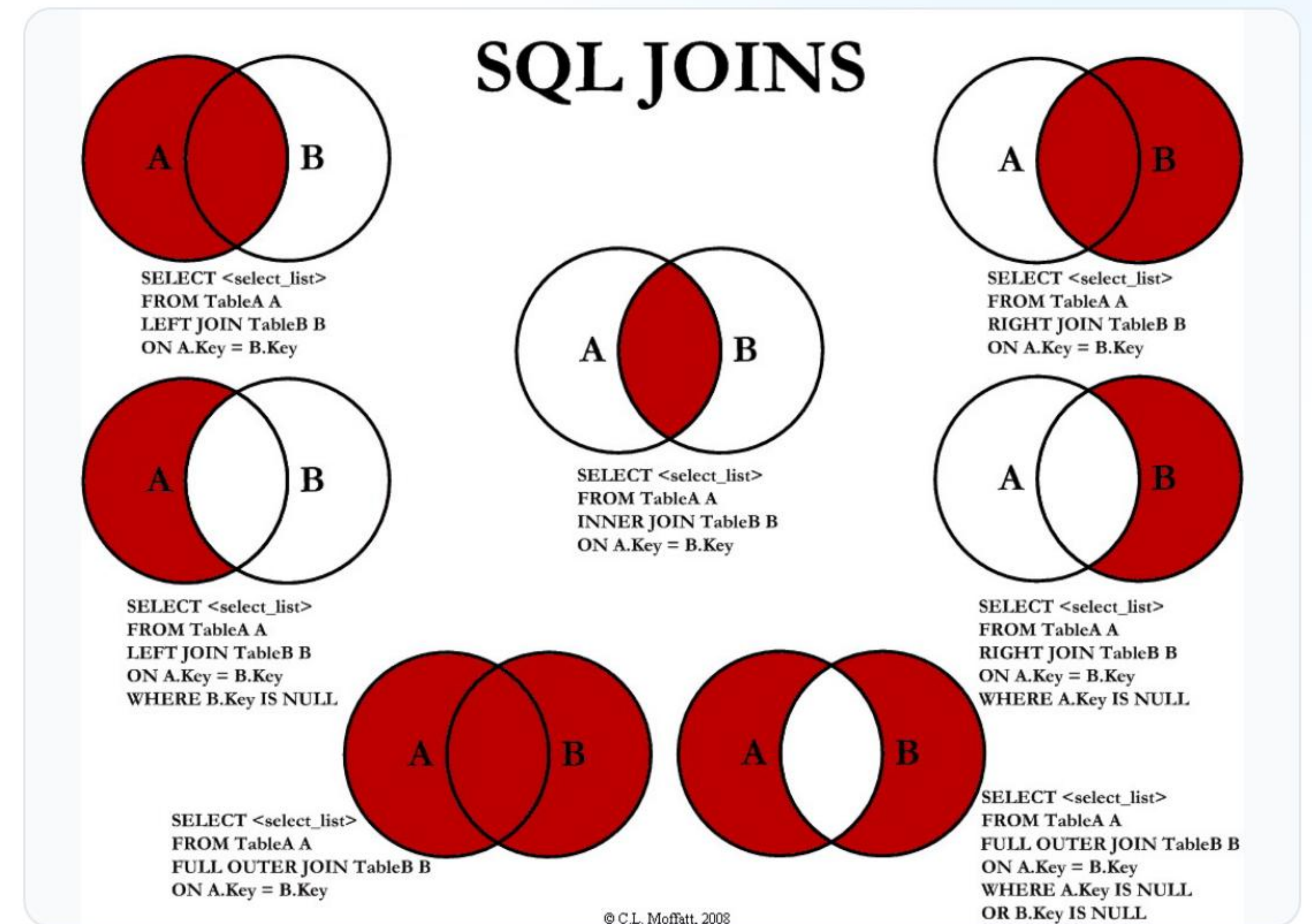
Data Joins Explained

What is a Join?

Joins are used to combine tables from the **same data source** based on a common field (a "key"). In Tableau, you perform joins on the Data Source page.

Common Types:

- **Inner Join:** Keeps only matching rows from both tables.
- **Left Join:** Keeps all rows from the left table and matching rows from the right table.

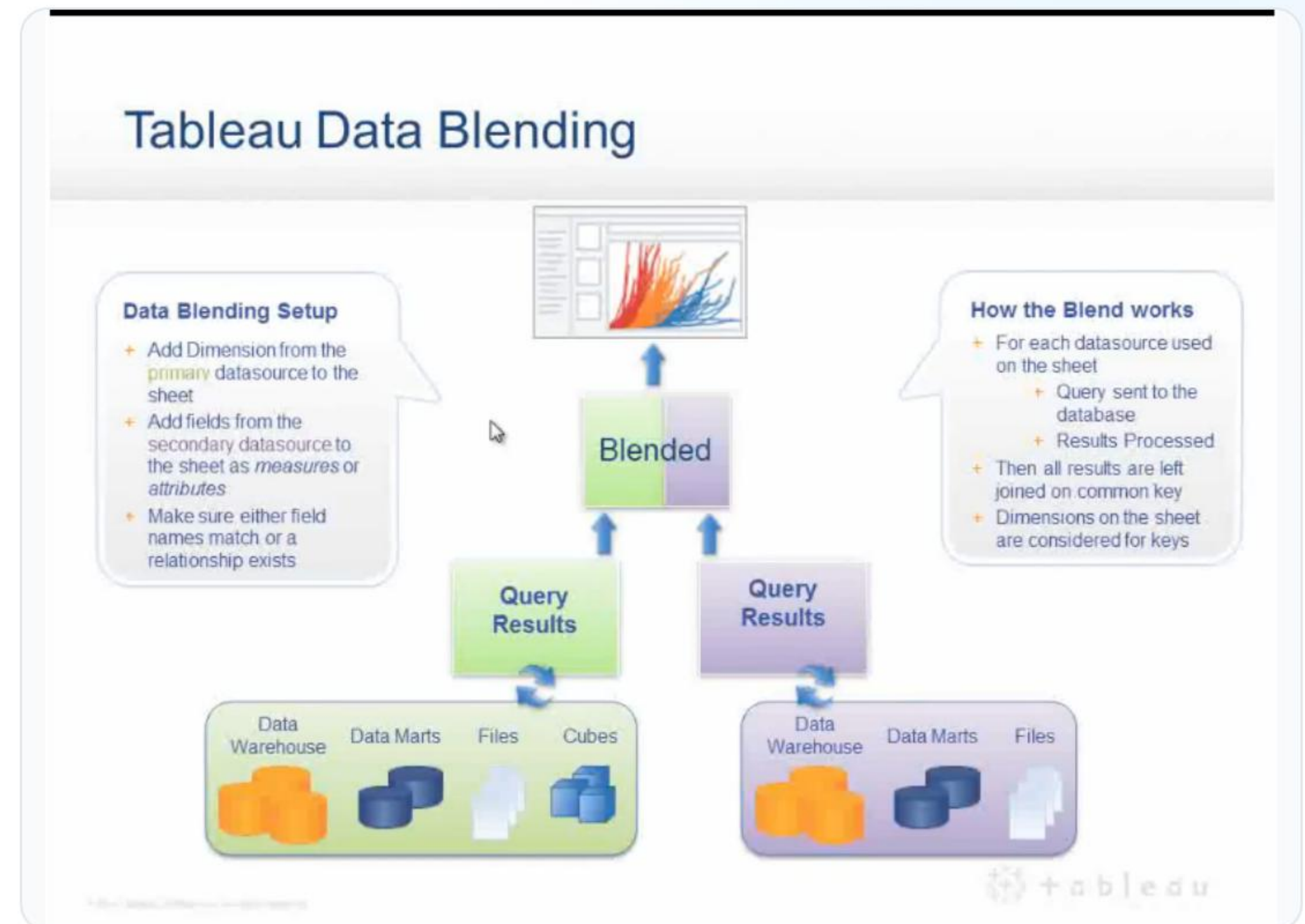


Data Blending Explained

What is a Data Blend?

Blending is used to combine data from **different data sources** (e.g., our two separate CSVs). Blending happens **after** aggregation.

- **Primary Source (Blue):** The first data source you use in a viz. Defines the level of detail.
- **Secondary Source (Orange):** The second source. Data is aggregated to match the primary's level.
- A "link" icon appears next to the common field.



Join vs. Blend: Key Differences

JOINS (Data-Level)

- Happens on the **Data Source** page.
- Combines tables *before* calculation.
- Creates a single, new "wide" table.
- Used for sources within the **same connection**.
- Can create row duplication if not careful.

BLENDING (Aggregate-Level)

- Happens on the **Worksheet**.
- Combines *after* aggregating data.
- Queries each source independently.
- Used for **different connections** (like our two CSVs).
- Secondary data is **always** aggregated.

Data Unions Explained

What is a Union?

A Union is used to **stack data** from multiple tables that have the same columns. It appends rows to each other.

Example: You have `Sales\_2024.csv` and `Sales\_2025.csv`. A union would combine them into one table with all rows from both years.

You can create a union by dragging a new table **under** the first table in the Data Source page.

Quiz: Connecting Data

Q1
Quiz Time

Question

You need to combine `hr\_analytics.csv` and `sales\_data.csv`. They are two separate files and do not have a common key. What is the **only** way to show data from both on the same dashboard?

A) Inner Join, **B)** Data Blend, **C)** Separate Dashboards, **D)** Union

Quiz: Solution

A: C
Solution

Answer: C) Separate Dashboards

This is a trick question! A **Data Blend** (B) requires a common linking field. Since our two files have no logical key, you cannot blend them. You must analyze them separately, likely on different dashboards or separate vizzes on the same dashboard that are not linked.

Mastering Calculated Fields

Beyond simple math: Logic, Aggregation, and LODs.

Logic: IF/THEN/ELSEIF

Conditional Logic

The **IF** function allows you to create custom categories based on your data. This is one of the most common tasks in data analysis.

Syntax:

IF [condition] **THEN** [value]

ELSEIF [condition 2] **THEN** [value 2]

ELSE [default value]

END

Calculated Field [IF THEN Classification Formula]

Name: IF THEN Classification Formula

Formula:

```
IF [Student GPA] >= 3 THEN "Advanced"
ELSEIF [Student GPA] >= 2 THEN "Average"
ELSE "Below Average" END
```

✓ The calculation is valid. Clear

Fields: All Enter Text to Search

- Case Numeric Grade Formula
- Class
- IF THEN Certificate Formula
- IF THEN Certificate Null Formula
- IF THEN Numeric Grade Formula
- IIF Certificate Formula
- IIF Numeric Grade Formula
- Letter Grade (No Nulls)
- Letter Grade (Nulls)
- Number of Records
- Student
- Student GPA

Parameters: All Enter Text to Search

Functions: Logical Enter Text to Search

- CASE
- IF
- IFNULL
- IIF
- ISDATE
- ISNULL
- ZN

IF <expr> THEN <then> [ELSEIF <expr2> THEN <then2> ...] [ELSE <else>] END

Tests a series of expressions returning the <then> value for the first true <expr>.

Example: IF Profit > 0 THEN 'Profitable' ELSEIF Profit = 0 THEN 'Break-even' ELSE 'Loss' END

OK Cancel Apply

Exercise: Salary Bands

Ex. 11

Calculation

Exercise: Create Salary Bands

Dataset: `hr_analytics.csv`

Task: Create a new Calculated Field named "Salary Band" that categorizes employees:

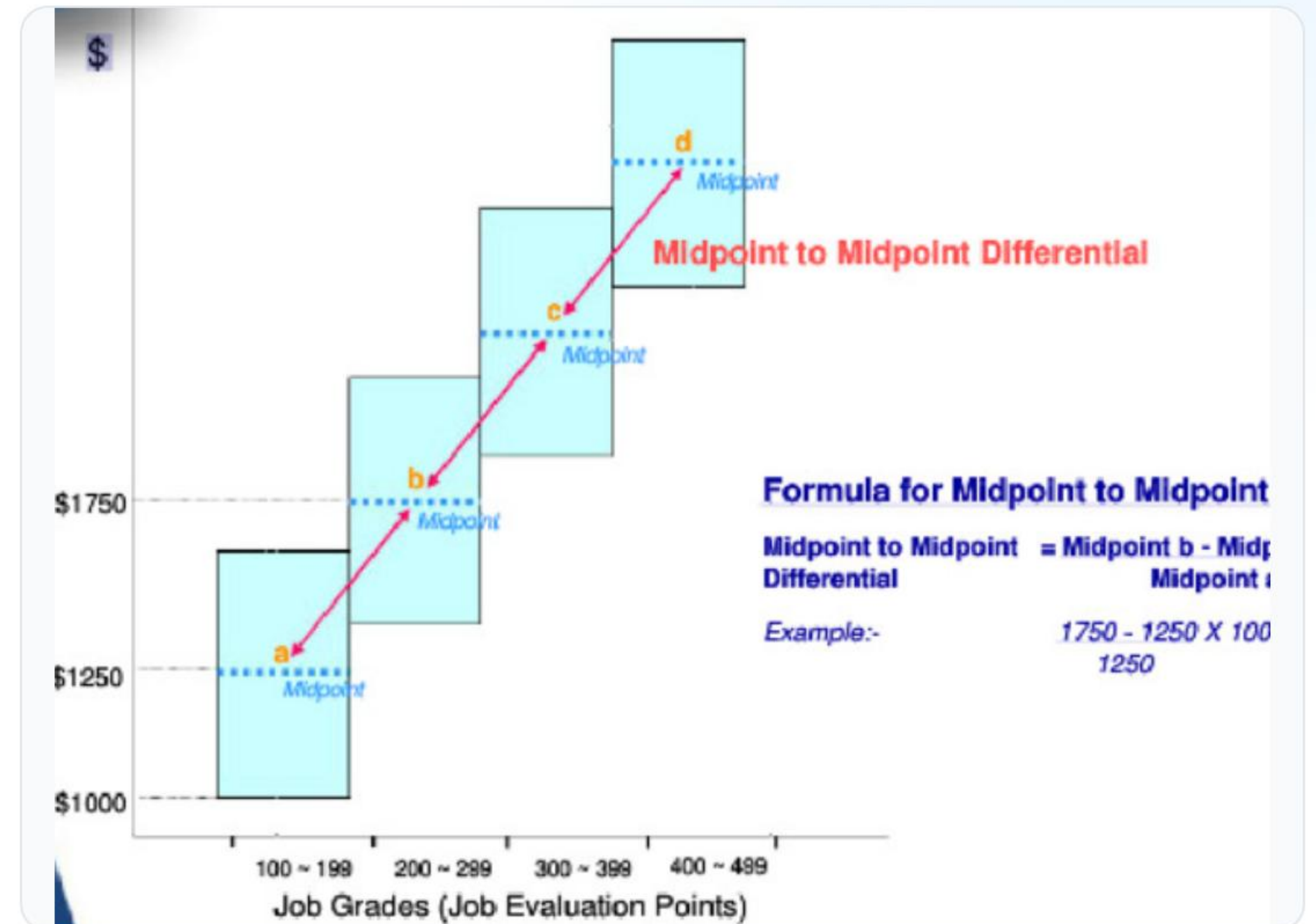
- $\leq 60,000$: "Entry-Level"
- $\leq 120,000$: "Mid-Level"
- $> 120,000$: "Senior-Level"

Solution: Salary Bands

Calculated Field

Your new dimension can now be used to slice your data, for example, to see the average performance rating by salary band.

```
IF [Salary] <= 60000 THEN "Entry-Level"  
ELSEIF [Salary] <= 120000 THEN "Mid-Level"  
ELSE "Senior-Level"  
END
```



Logic: CASE Statements

What is a CASE Statement?

A **CASE** statement is a cleaner, more efficient way to write an **IF** statement when you are evaluating a **single field** against multiple exact values.

Syntax:

```
CASE [field]
```

```
WHEN "value 1" THEN "result 1"
```

```
WHEN "value 2" THEN "result 2"
```

```
ELSE "default result"
```

```
END
```

Calculated Field [CASE Numeric Grade Formula]

Name: CASE Numeric Grade Formula

Formula:

```
CASE [Letter Grade (No Nulls)]  
WHEN "A" THEN 4  
WHEN "B" THEN 3  
WHEN "C" THEN 2  
WHEN "D" THEN 1  
ELSE 0 END
```

✓ The calculation is valid.

Fields: All

Parameters: All

Functions: Logical

Enter Text to Search

Abc Class
Abc IF THEN Certificate Formula
Abc IF THEN Certificate Null Formula
Abc IF THEN Numeric Grade Formula
Abc IIF Certificate Formula
Abc IIF Numeric Grade Formula
Abc Letter Grade (No Nulls)
Abc Letter Grade (Nulls)
Abc Number of Records
Abc Student

Enter Text to Search

Enter Text to Search

Logical
CASE
IF
IFNULL
IIF
ISDATE
ISNULL
ZN

CASE <expr> WHEN <value1> THEN <return1> ... [ELSE <else>]
END

Finds the first <value> that matches <expr> and returns the corresponding <return>.

Example: CASE RomanNumeral WHEN 'I' THEN 1 WHEN 'II' THEN 2 ELSE 3
END

OK Cancel Apply

Exercise: Performance Names

Ex. 12

Calculation

Exercise: Readable Performance

Dataset: `hr_analytics.csv`

Task: The `Performance_Rating` field is good, but HR wants a new field "Performance Group" for a report.

Use a `CASE` statement to group "Excellent" and "Good" as "Exceeds Expectations" and "Satisfactory" / "Needs Improvement" as "Meets or Below".

Solution: Performance Names

```
CASE [Performance_Rating]
WHEN "Excellent" THEN "Exceeds Expectations"
WHEN "Good" THEN "Exceeds Expectations"
WHEN "Satisfactory" THEN "Meets or Below"
WHEN "Needs Improvement" THEN "Meets or Below"
END
```

Level of Detail (LOD) Expressions

“

This is the most important concept in Tableau. LODs allow you to calculate values at a different level of detail than your visualization.

”

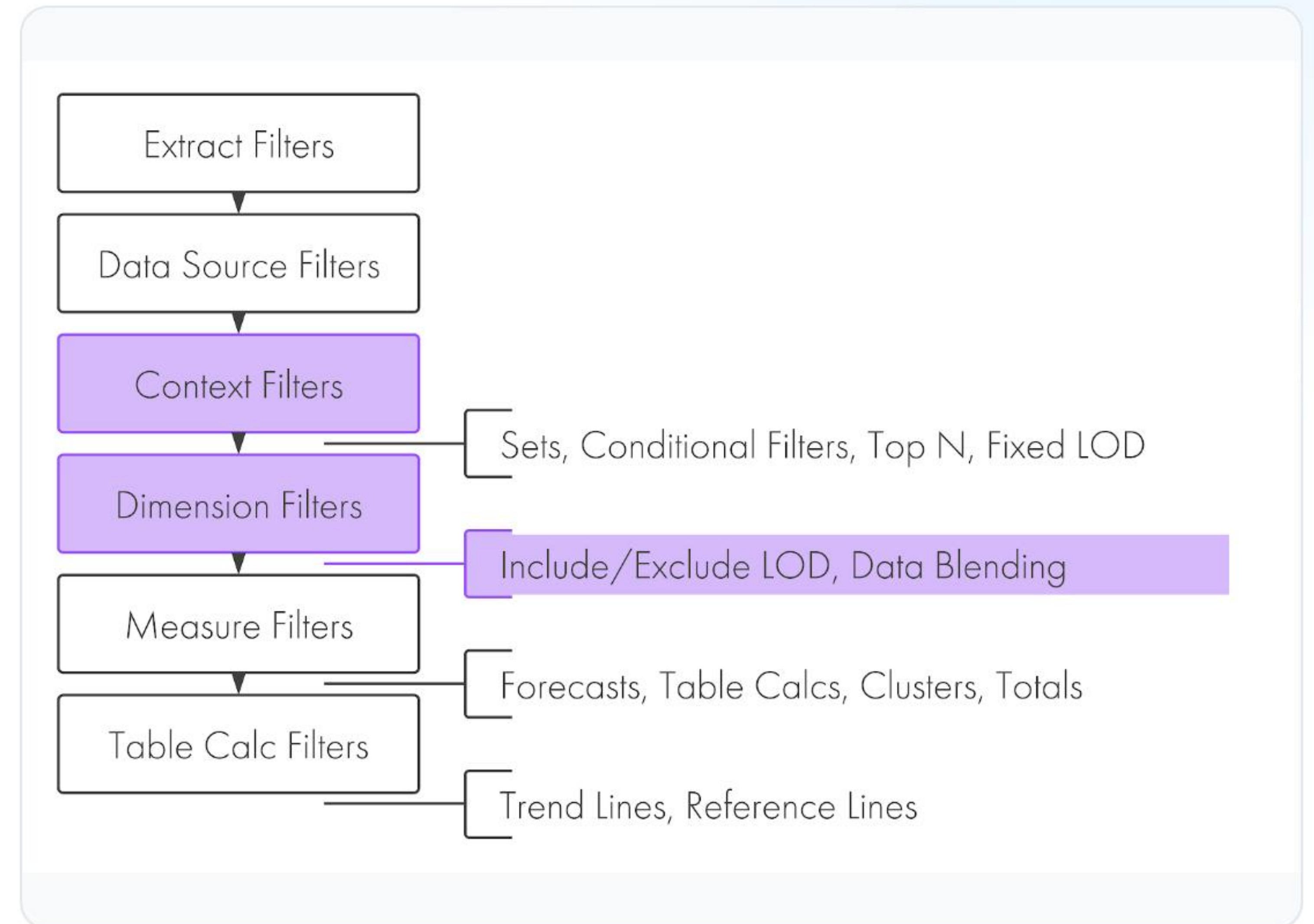
— *Every Tableau Expert*

LOD Type 1: FIXED

The `FIXED` LOD

This is the most powerful LOD. It computes an aggregate value for a specific dimension, **ignoring all other dimensions and most filters in the view.**

Use Case: "What is the total sales for this customer?" (and show that same total next to every *order* from that customer).



Exercise: Avg. Dept Salary

Ex. 13

FIXED LOD

Exercise: Department Salary Average

Dataset: `hr_analytics.csv`

Task: Create a calculated field "Avg Dept Salary" that shows the average salary for each employee's *entire department*.

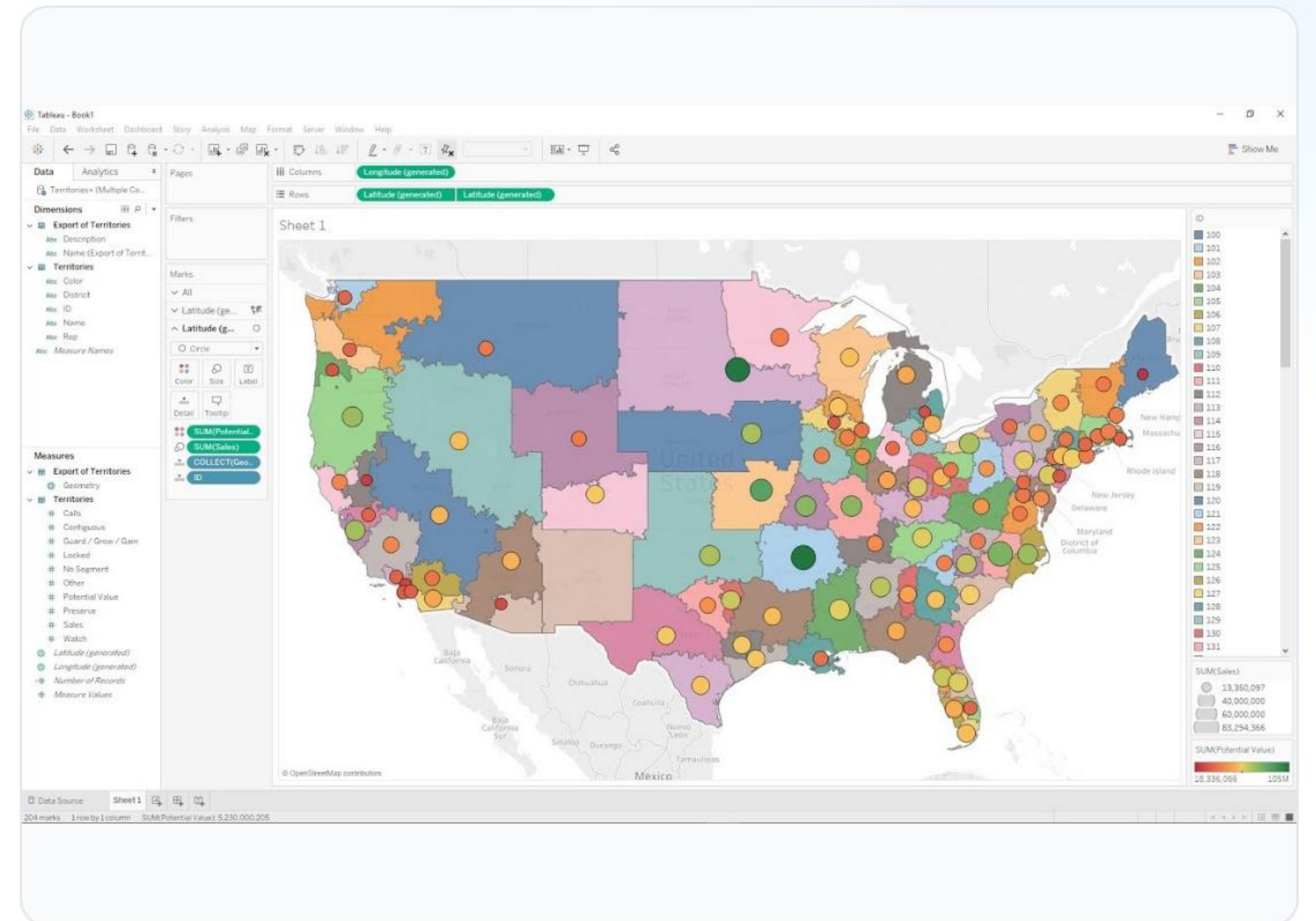
Test: Put `Employee_Name` , `Salary` , and your new `[Avg Dept Salary]` field in a table. The "Avg Dept Salary" value should be the same for all employees in the same department.

Solution: Avg. Dept Salary

LOD Calculation

This calculation first goes to the data, finds each `[Department]`, computes the `AVG([Salary])`, and then "fixes" that value to the department, regardless of what else is in the view (like `Employee_Name`).

```
{ FIXED [Department] : AVG([Salary]) }
```



LOD Types 2 & 3: INCLUDE / EXCLUDE

INCLUDE

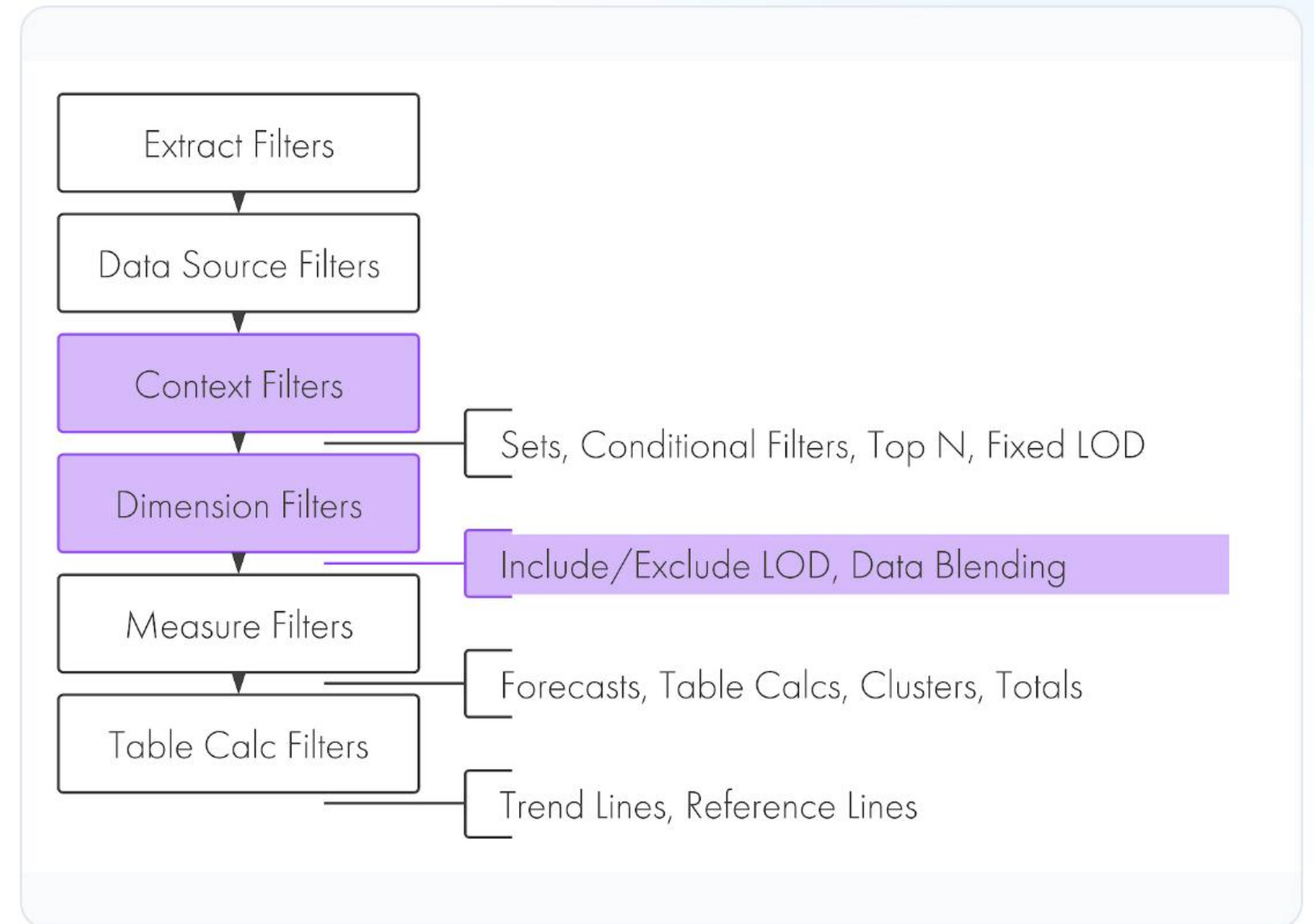
Computes an aggregate at a **more granular** level than the view.
"For each Region in my view, what was the average sale *per customer*?"

```
{ INCLUDE [Customer_ID] : SUM([Sales_Amount]) }
```

EXCLUDE

Computes an aggregate at a **less granular** level than the view.
"For each Department AND Job Title in my view, what was the average salary for the *entire department*?"

```
{ EXCLUDE [Job_Title] : AVG([Salary]) }
```



A Note on Context Filters

"

By default, `FIXED` LODs ignore your filters. If you want a `FIXED` LOD to be affected by a filter, you must "**Add to Context**" by right-clicking the filter.

"

This is a critical, advanced concept for dashboard performance.

Quiz: LODs & Calcs

Q2

Quiz Time

Question

You want to calculate the total sales for each **Region** , and you want this value to remain the same even if a user filters by **City** . Which calculation should you use?

A) `SUM([Sales_Amount])` , B) `{ INCLUDE [Region] : SUM([Sales_Amount]) }` , C) `{ FIXED [Region] : SUM([Sales_Amount]) }`

Quiz: Solution

A: C
Solution

Answer: C) `FIXED`

```
{ FIXED [Region] : SUM([Sales_Amount]) }
```

A standard `SUM([Sales_Amount])` (A) would be affected by the City filter. The `FIXED` LOD (C) calculates *before* the City filter is applied, giving you the total for the entire region.

Dynamic Visualizations

Using Sets and Parameters for Interactivity.

What is a Set?

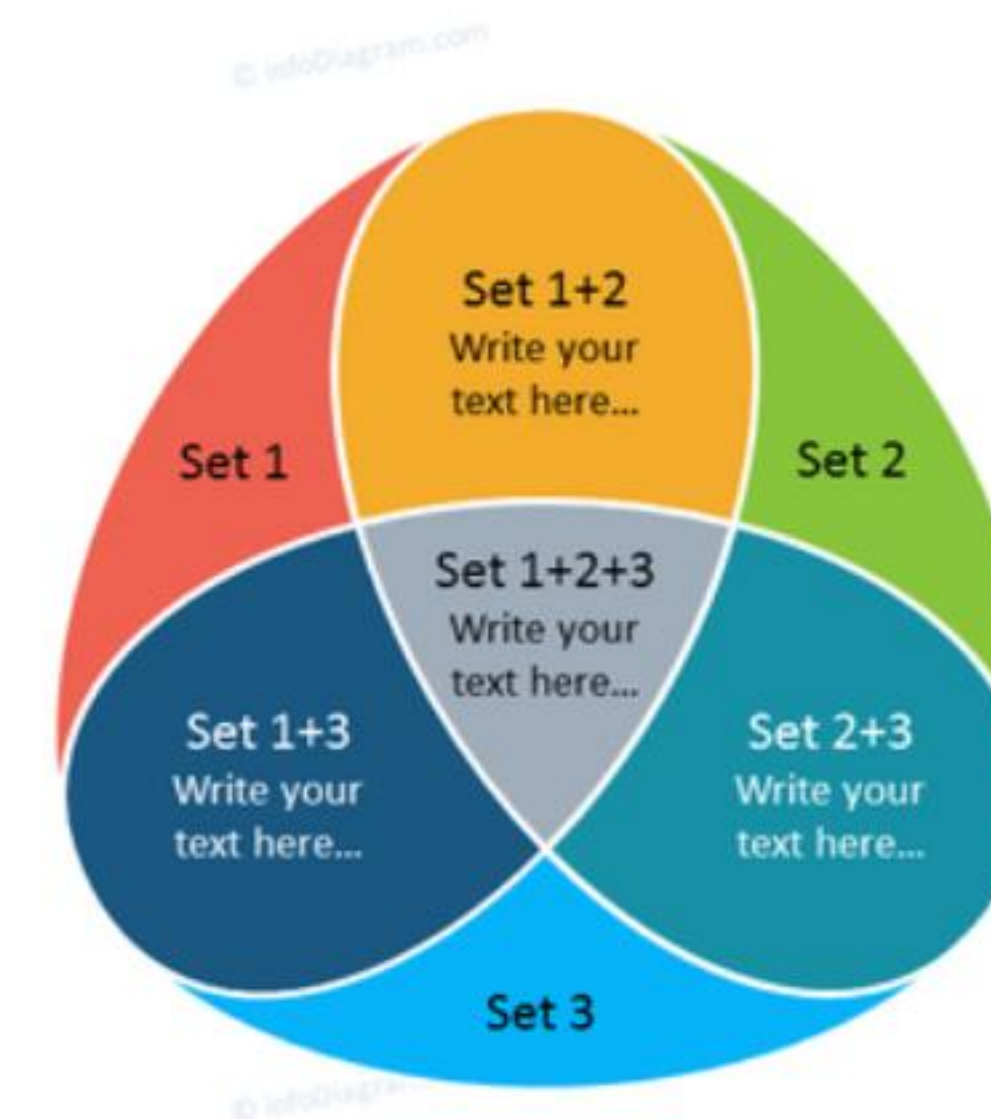
Sets create an "IN" / "OUT" group.

A set is a custom field that defines a subset of your data. Think of it as a binary grouping: members are either **IN** the set or **OUT** of the set.

- **Static Set:** You manually select members (e.g., "Spain", "France", "Germany" in a "EU" set).
- **Dynamic Set:** You define a *condition* (e.g., Top 10 Customers by Profit). This set updates automatically as data changes.

Three Set Euler Diagram

with Four Shared Groups & Venn-like Intersections



Category 1

Write your text here. Fill the textbox with your description. Add description of an item.

Category 2

Write your text here. Fill the textbox with your description. Add description of an item.

Category 3

Write your text here. Fill the textbox with your description. Add description of an item.

Category 1+2

Write your text here. Fill the textbox with your description. Add description of an item.

Category 1+3

Write your text here. Fill the textbox with your description. Add description of an item.

Category 2+3

Write your text here. Fill the textbox with your description. Add description of an item.

Category 1+2+3

Write your text here. Fill the textbox with your description. Add description of an item.

Get these slides & icons at www.infoDiagram.com

Exercise: Top 10 Customers

Ex. 14

Sets

Exercise: Find Your Top Customers

Dataset: `sales_data.csv`

Task: Create a **Dynamic Set** named "Top 10 Customers by Sales".

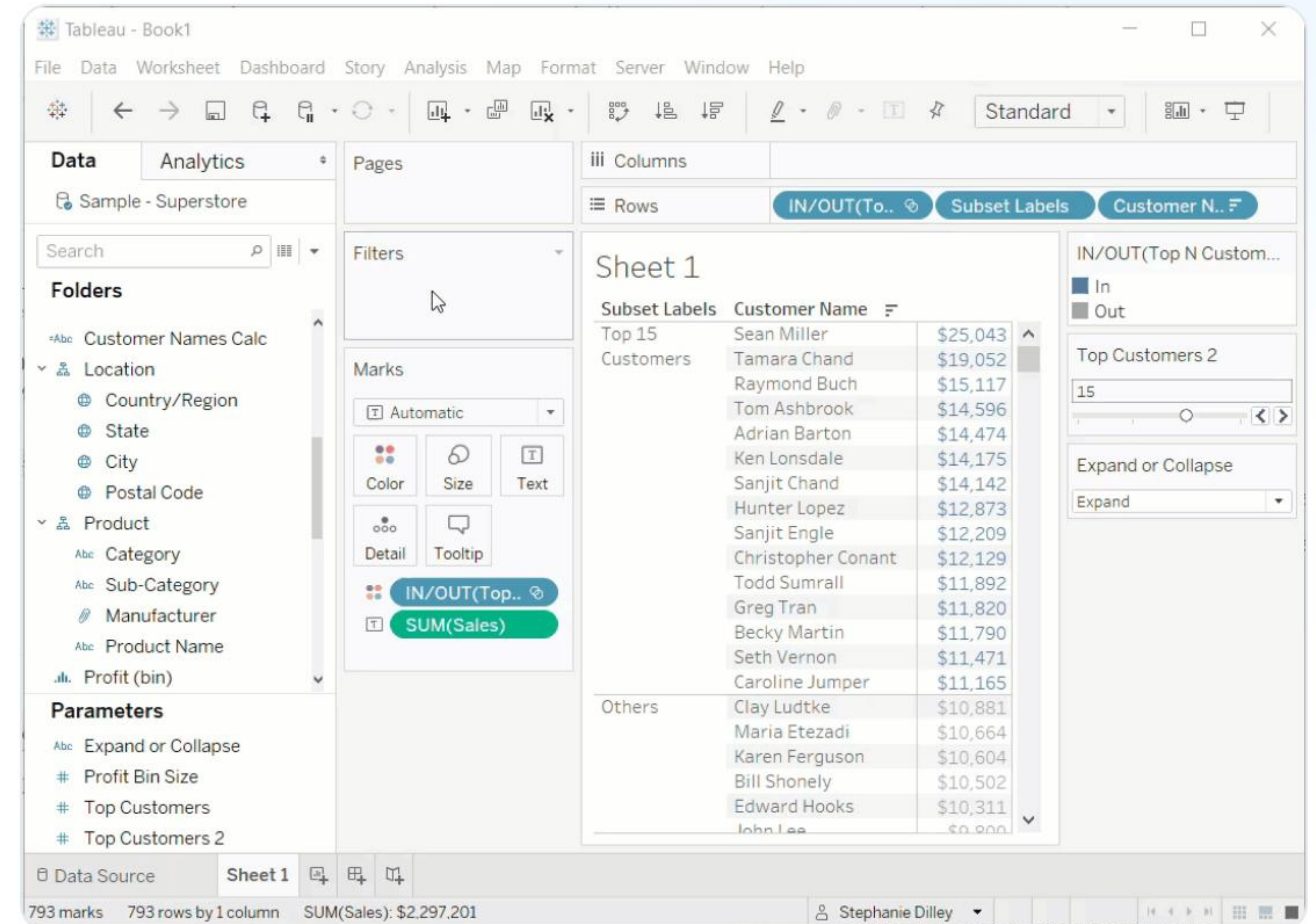
Test: Create a bar chart of `Customer_Name` by `SUM(Sales_Amount)`. Drag your new set to the **Color** shelf. This should highlight your Top 10 customers.

Solution: Top 10 Customers

Creating the Set

1. Right-click **Customer_Name** in the Data Pane > Create > Set.
2. Name it "Top 10 Customers by Sales".
3. Go to the "Top" tab.
4. Select "By field": Top 10 by **Sales_Amount** (Sum).

Using this set on Color instantly segments your most valuable customers.



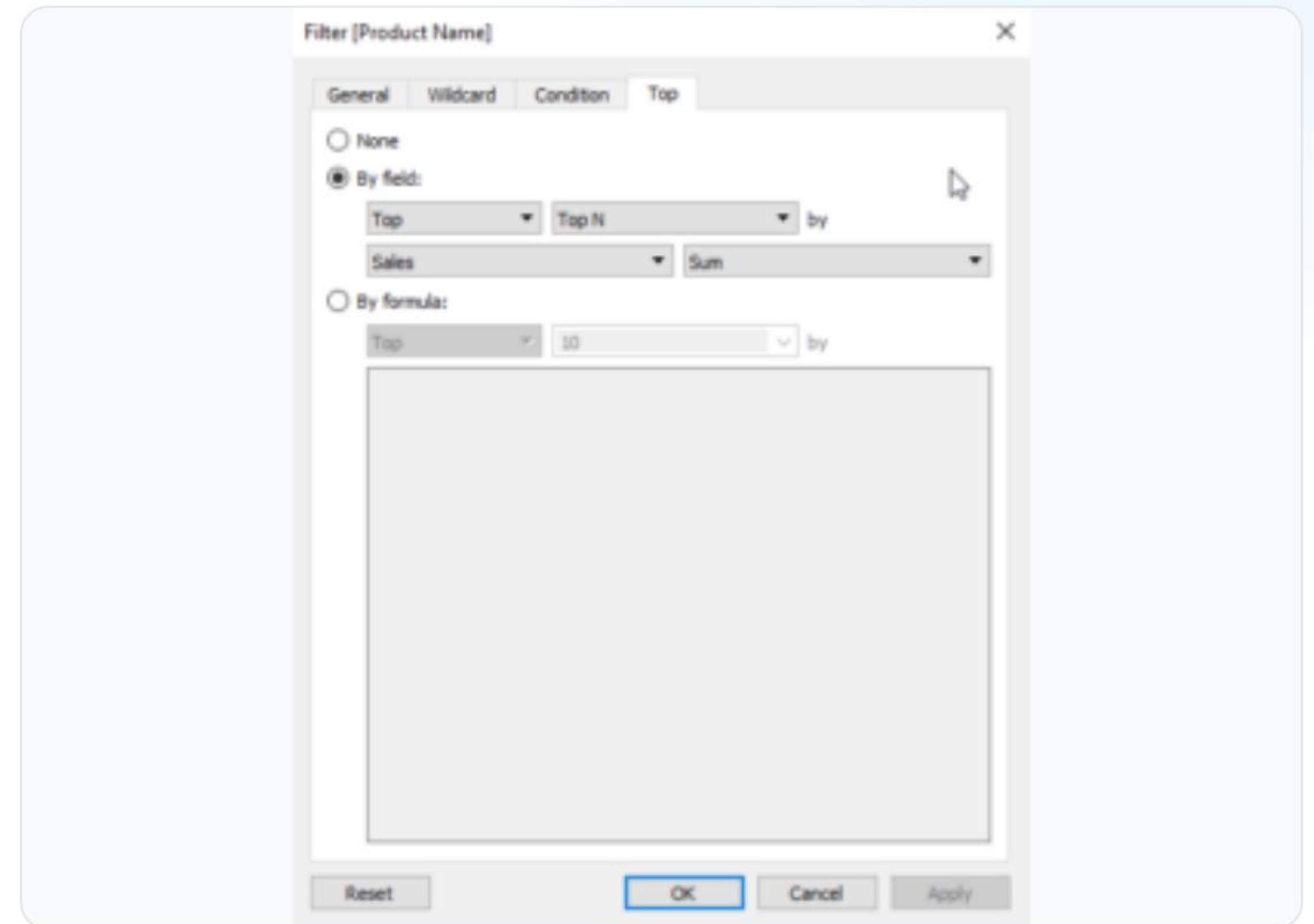
What is a Parameter?

A User-Controlled Variable

A Parameter is a workbook variable that a user can change. It doesn't do anything on its own; it must be used inside a **Calculation, Filter, or Reference Line**.

This is the key to "What-If" analysis.

Example: A user types "10" into a parameter box, and your "Top N" filter updates to show the Top 10.



Exercise: "Top N" Parameter

Ex. 15

Parameters

Exercise: Make Your Set Dynamic

Task: Let's combine our Set with a Parameter.

1. Create a new **Parameter** called "Top N" (Integer, Range 5 to 20).
2. Edit your "Top 10 Customers" set.
3. In the "Top" tab, change the dropdown from "10" to your new **[Top N]** parameter.
4. Show the Parameter control on your sheet.

Solution: "Top N" Parameter

Dynamic Control

You now have a fully interactive chart. When the user moves the "Top N" slider from 5 to 10 to 20, the set re-calculates, and the colors on your chart update instantly.

This is a core skill for professional dashboards.



Exercise: "What-If" Discount

Ex. 16

What-If

Exercise: "What-If" Analysis

Dataset: `sales_data.csv`

Task: The sales team wants to know: "What would our profit look like if we applied an additional discount?"

1. Create a Parameter "What-If Discount" (Float, Range 0.0 to 0.2, step 0.01).
2. Create a Calc. Field "What-If Profit":
`[Profit] - ([Sales_Amount] * [What-If Discount])`

Advanced Chart Types

Telling stories with Maps, Dual-Axis charts, and Treemaps.

Geographic Maps

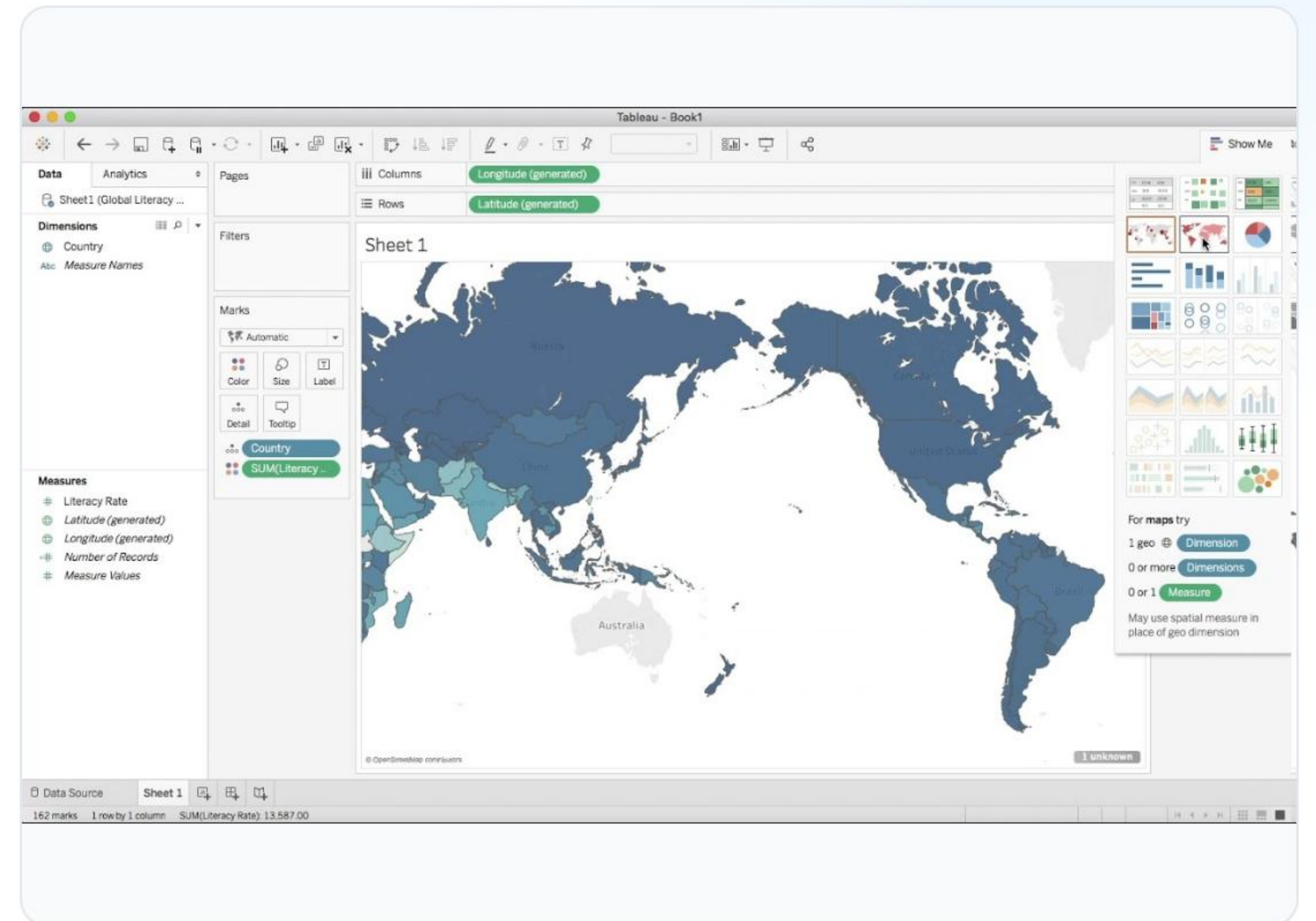
Visualizing Spatial Data

Tableau excels at maps. If your data has geographic fields (like Country, City), Tableau will auto-generate latitude and longitude.

Simply double-click a geo-field (like **Country**) to create a map.

Common Uses:

- **Color:** Use a measure like **AVG(Profit)** on Color.
- **Size:** Use a measure like **SUM(Sales_Amount)** on Size.



Exercise: Sales & Profit Map

Ex. 17

Maps

Exercise: Global Sales Map

Dataset: `sales_data.csv`

Task: Create a world map that shows:

1. `Country` as the level of detail.
2. `SUM(Sales_Amount)` on the **Size** shelf.
3. `AVG(Profit)` on the **Color** shelf.

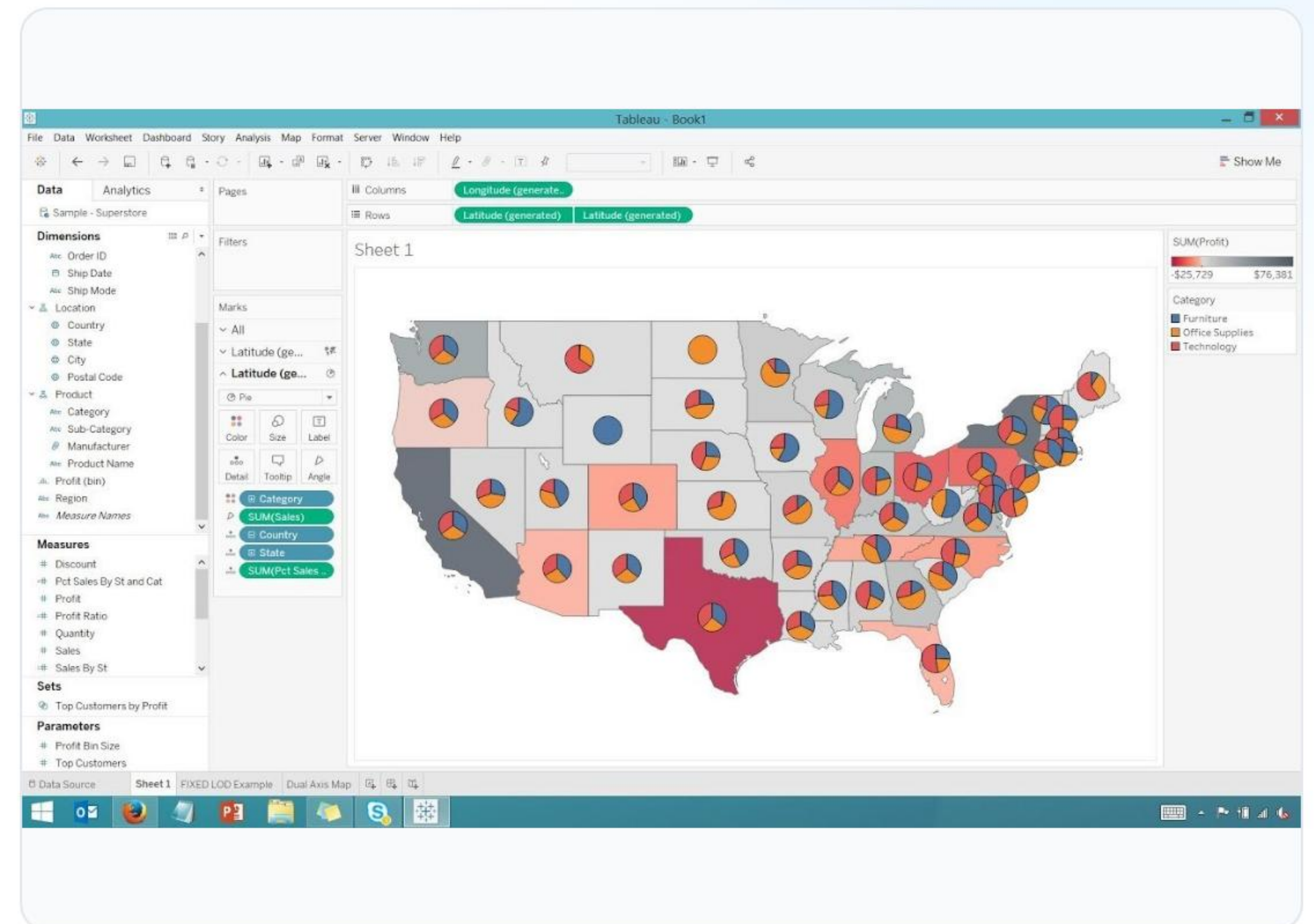
Hint: Edit the colors to be a diverging Red-to-Green palette to show losses/gains.

Solution: Sales & Profit Map

Global Insights

This single map tells a rich story. You can immediately see that while some countries have large sales (big circles), they may be unprofitable (red circles).

This is a classic "analysis-ready" visualization that provides immediate insight.



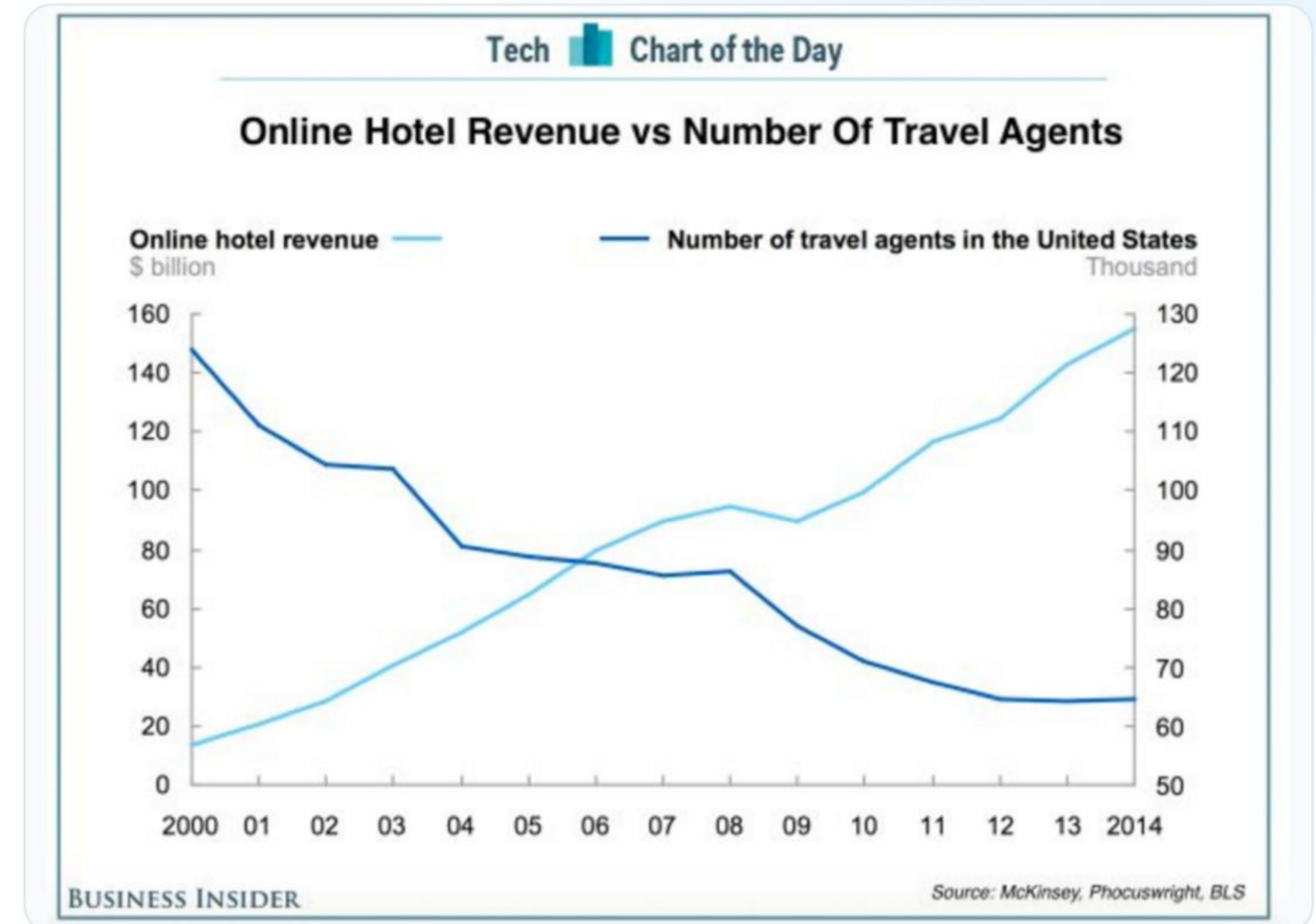
Dual-Axis Charts

Comparing Two Measures

A dual-axis chart places two measures on two separate Y-axes. This is perfect for comparing two measures with **different scales**, like **Sales (in \$1,000s)** and **Profit (in \$100s)** over time.

How to Build:

1. Drag **Date** to Columns.
2. Drag **SUM(Sales_Amount)** to Rows.
3. Drag **SUM(Profit)** to the *opposite* side of the view until you see a green line, then drop.
4. **Crucial:** Right-click the second axis and select "**Synchronize Axis**" if the scales should align (or leave them independent if not).



Exercise: Sales vs. Profit Trend

Ex. 18

Dual-Axis

Exercise: Sales & Profit Over Time

Dataset: `sales_data.csv`

Task: Create a dual-axis chart.

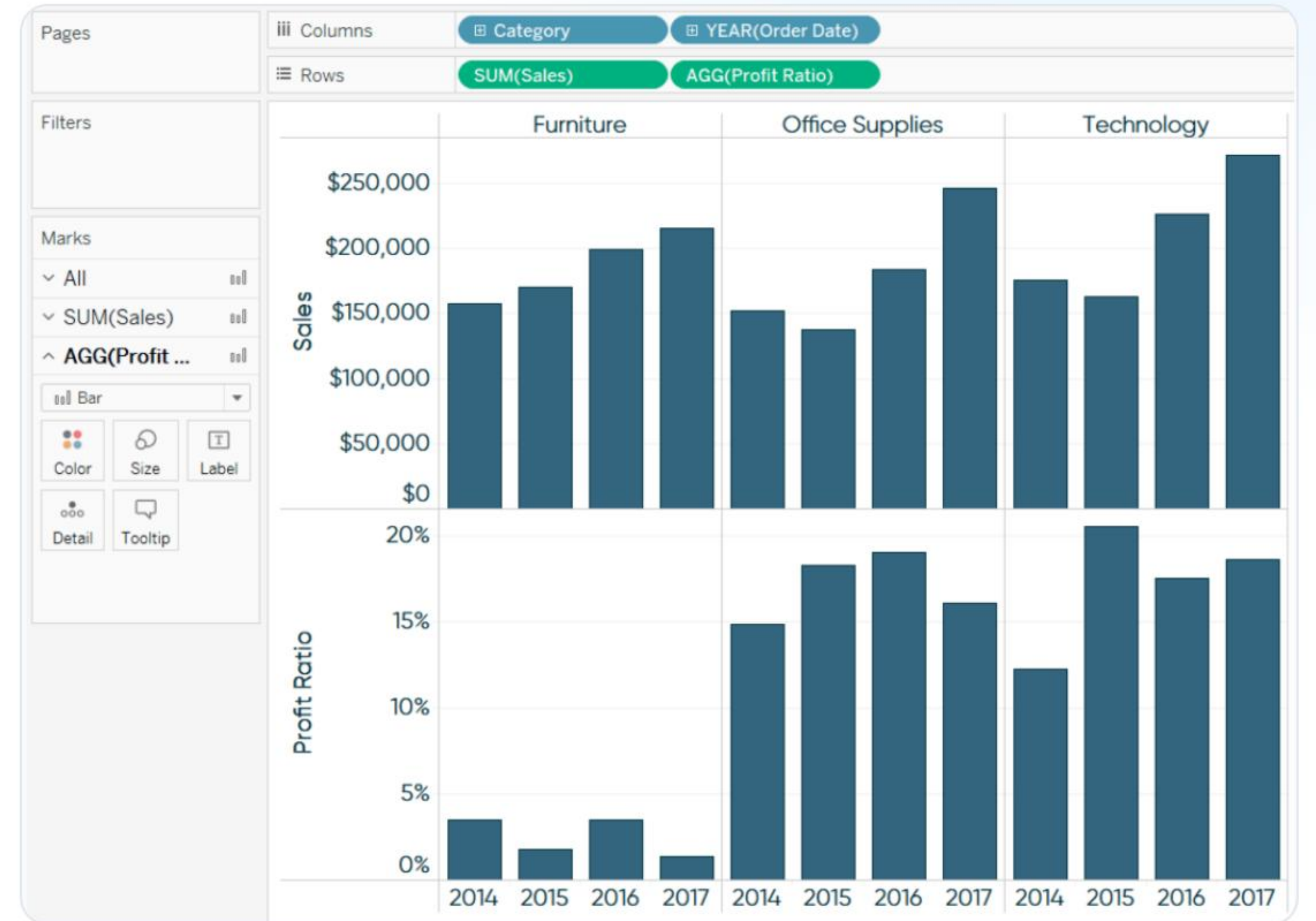
1. X-Axis: Continuous `MONTH(Date)` .
2. Y1-Axis: `SUM(Sales_Amount)` (as a Bar).
3. Y2-Axis: `SUM(Profit)` (as a Line).

Hint: Use the "Marks" card to change the chart type for each measure independently.

Solution: Sales vs. Profit Trend

Combined Insights

This "combo chart" is a powerful way to see the relationship between two measures. You can now see if months with high sales also resulted in high profit.



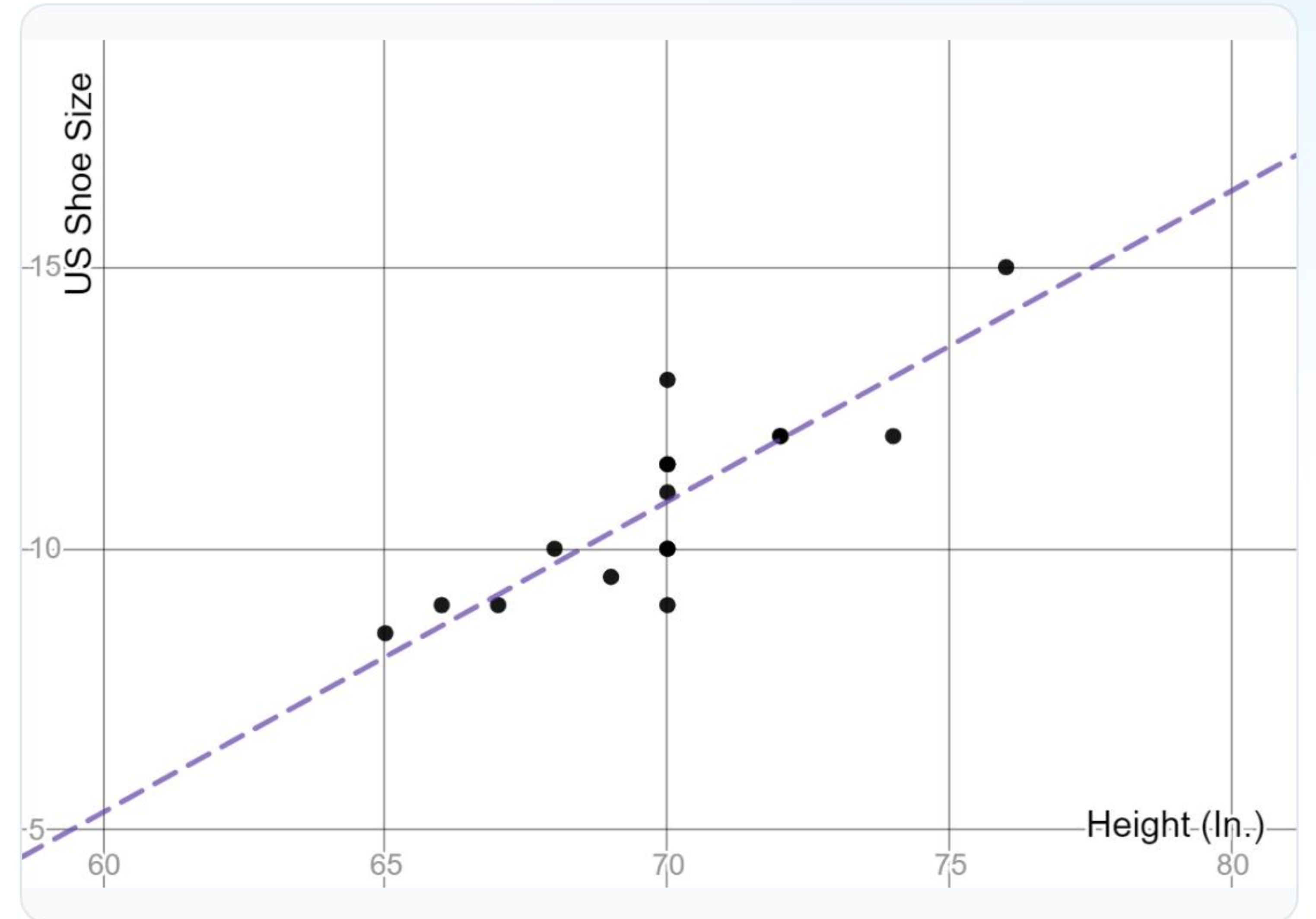
Scatter Plots

Finding Correlations

A scatter plot is the standard way to visualize the relationship (or correlation) between two different measures.

How to Build:

1. Drag one **Measure** to Columns.
2. Drag another **Measure** to Rows.
3. Tableau will create one dot. To break it down, drag a **Dimension** (like **Employee_ID** or **Customer_Name**) to the **Detail** shelf.
4. Go to the **Analytics** pane and drag a **Trend Line** onto the viz.



Exercise: Salary vs. Age

Ex. 19

Scatter Plot

Exercise: Age & Salary Correlation

Dataset: `hr_analytics.csv`

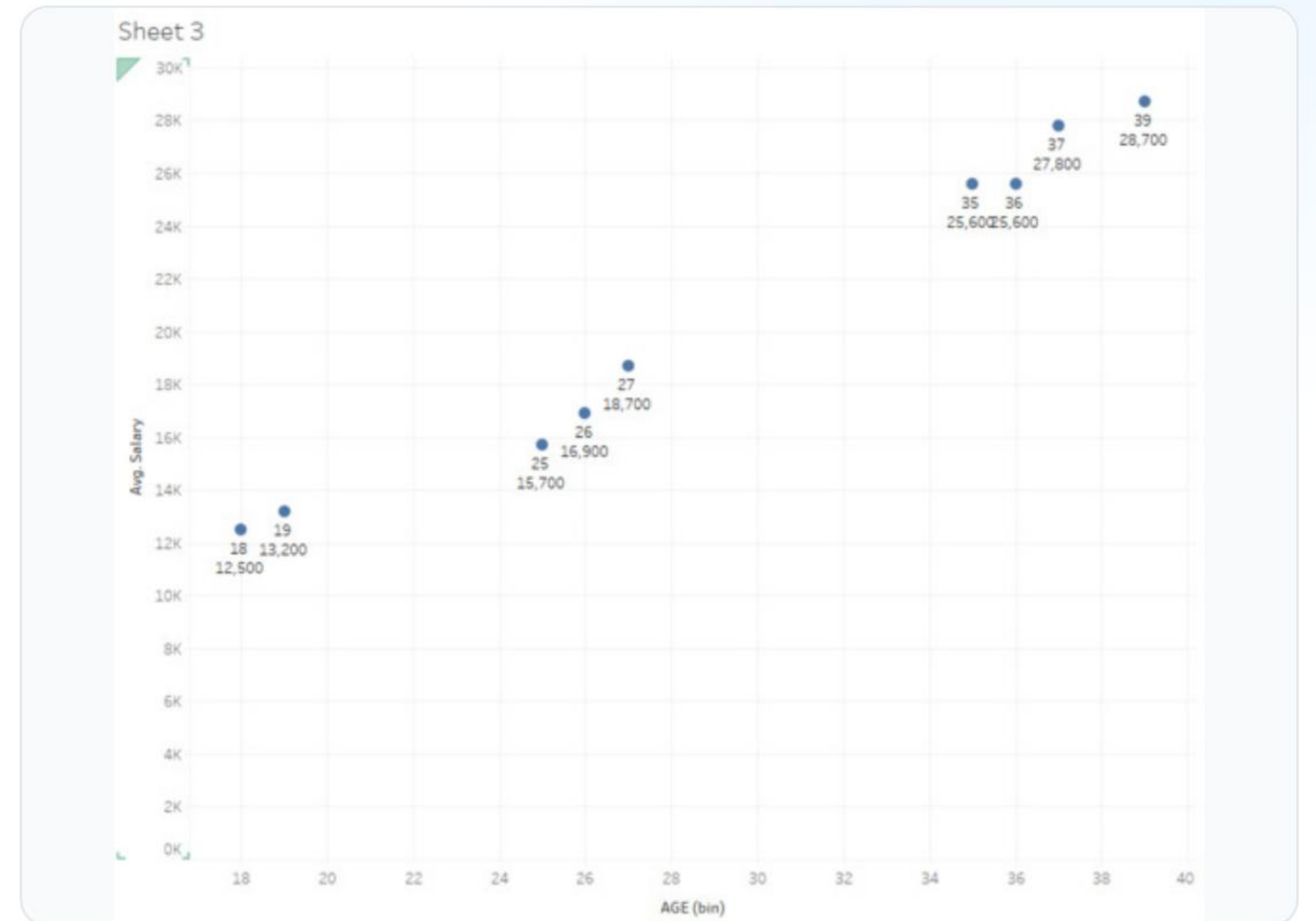
Task: Does salary increase with age at this company?

1. X-Axis: `Age` (as a Dimension).
2. Y-Axis: `Salary` (as a SUM).
3. Detail: `Employee_ID` .
4. Color: `Department` .
5. Add a **Trend Line**.

Solution: Salary vs. Age

Correlation Analysis

This scatter plot shows every employee as a dot. The trend line (and its R-squared value) will tell you how strong the correlation is. You can also spot outliers (e.g., a young employee with a very high salary).



Other Key Chart Types

Treemaps

Excellent for showing hierarchical data as a part-of-a-whole. The size of the rectangle represents a measure (like Sales), and the color can show a second measure (like Profit).

Use: Drag `Product_Category` to Label, `SUM(Sales)` to Size, and `AVG(Profit)` to Color.

Histograms

Shows the *distribution* of a single measure. How many employees are in each "Age" bucket? How many orders are "small" vs. "large"?

Use: Select a measure (like `Salary`), click "Show Me", and select Histogram. Tableau creates the bins for you.

Telling the Full Story

Advanced Dashboards, Actions, and Case Studies.

Dashboard Design Principles

From Charts to Insights

A dashboard is not just a collection of charts. It's a guided story that answers business questions.

- **Z-Layout:** Place your most important info (KPIs) in the top-left. Place details (tables) in the bottom-right.
- **Tiled vs. Floating:** Use **Tiled** containers (Vertical/Horizontal) for a responsive design. Use **Floating** only for small items like titles or logos.
- **Less is More:** Don't overcrowd. Use 3-4 visualizations that work together.

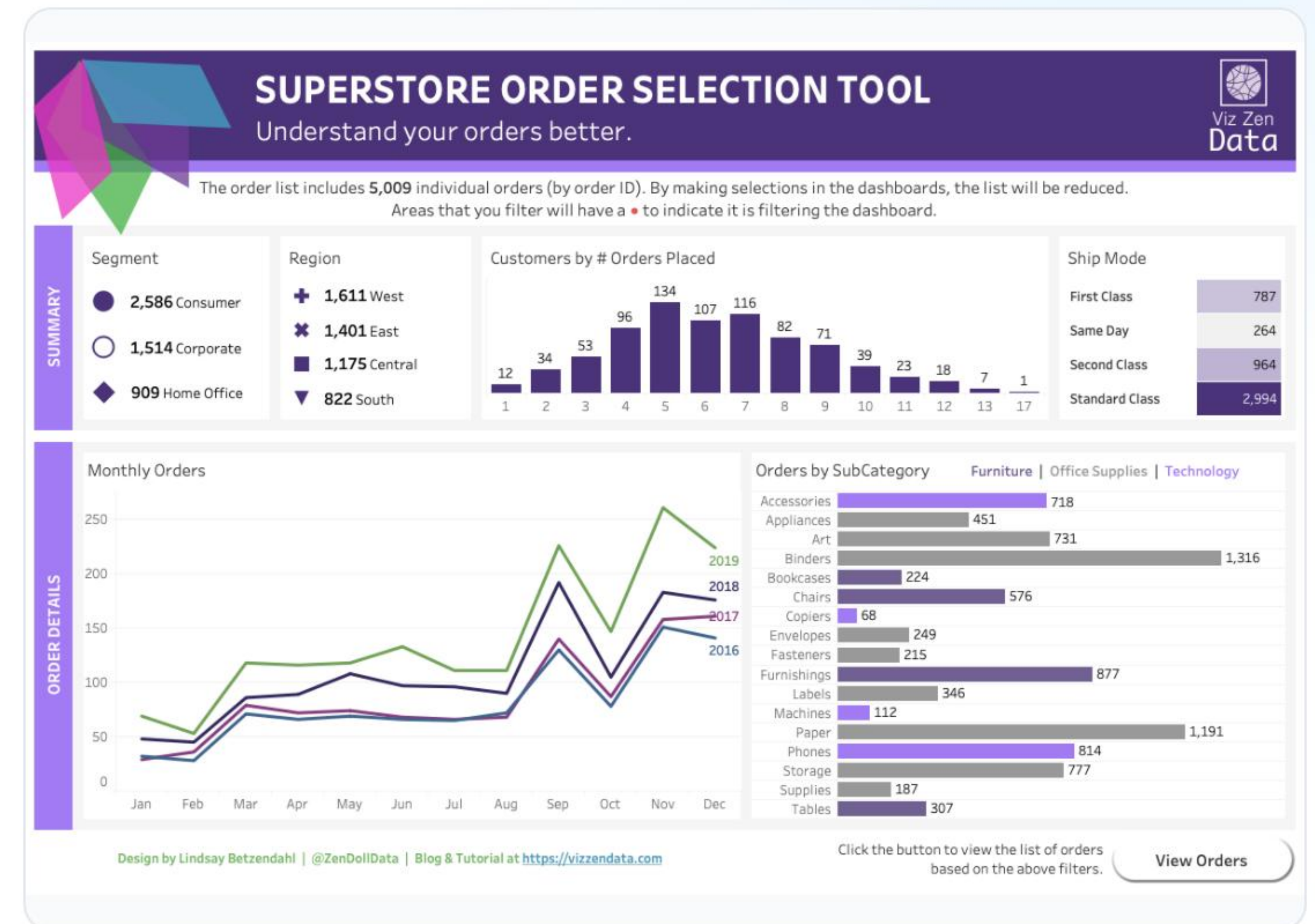


Dashboard Actions (Review)

Making it Interactive

Actions (found in the "Dashboard" menu) are what bring your dashboard to life. They connect your worksheets.

- **Filter (Most Common):** Use one sheet as a filter for others. (e.g., Click a **Region** on a map to filter a bar chart).
- **Highlight:** Click a **Department** to *highlight* those employees in a scatter plot, while graying out the rest.
- **URL:** Click a **Customer_Name** to open their Salesforce page in a browser.



Case Study 1: HR Attrition Analysis

The Business Question

"Why are our employees leaving? The CHRO needs to understand the key drivers of attrition to propose a retention strategy."

Your Goal

Build an interactive dashboard using `hr_analytics.csv` that analyzes the `Attrition` field against other key dimensions.



Employee Attrition Icon with Down Arrow

- 01 **Text Here**
This slide is 100% editable. Adapt it to your need and capture your audience's attention.
- 02 **Text Here**
This slide is 100% editable. Adapt it to your need and capture your audience's attention.
- 03 **Text Here**
This slide is 100% editable. Adapt it to your need and capture your audience's attention.

Case Study 1: Task

CS-1 HR Attrition

Create 3 Worksheets

1. **Overall Attrition Rate (KPI):** A single number showing the % of employees who have "Yes" for Attrition .
2. **Attrition by Dept (Bar):** A bar chart of Department vs. Attrition Rate.
3. **Attrition by Salary Band (Bar):** A bar chart of your "Salary Band" calc vs. Attrition Rate.

Case Study 1: Attrition Rate Calc

Calculated Field

You need a new measure for "Attrition Rate".

```
// Calc: Attrition Count  
IF [Attrition] = "Yes" THEN 1 ELSE 0 END
```

```
// Calc: Attrition Rate  
SUM([Attrition Count]) / COUNTD([Employee_ID])
```

Remember: Set the default number format for "Attrition Rate" to Percentage!

KPI Sheet

On a new sheet, drag your "Attrition Rate" measure to "Text".

Click "Text" on the Marks card, click the "...", and edit the text to be large and clear. Add a title to the sheet.

Case Study 1: Dashboard

Putting It Together

Assemble your three worksheets (KPI, Dept Chart, Salary Band Chart) onto one dashboard.

Add a **Filter Action** so that clicking on a **Department** in the bar chart filters the "Attrition by Salary Band" chart.

Insights

Your dashboard should reveal that Attrition isn't uniform. It's likely concentrated in specific departments (like Sales) and at specific salary levels (like Entry-Level).

Case Study 2: Sales Profitability

The Business Question

"Our sales are high, but our profits are not. The VP of Sales needs to know where we are losing money. Is it certain products, regions, or discounts?"

Your Goal

Build an interactive dashboard using `sales_data.csv` that analyzes the relationship between `Profit`, `Discount`, and `Product_Category`.



Case Study 2: Task

CS-2

Sales Profit

Create 3 Worksheets

1. **Profit Map (Map):** A map of `Country` , with `AVG(Profit)` on Color (use a Red-Green diverging palette).
2. **Profit vs. Discount (Scatter):** A scatter plot of `AVG(Profit)` vs. `AVG(Discount)` , with `Product_Category` on Color.
3. **Profit by Category (Bar):** A simple bar chart of `Product_Category` vs. `SUM(Profit)` .

Case Study 2: Profit vs. Discount

Key Analysis: The Scatter Plot

This is the most important chart for this case study. By plotting **AVG(Discount)** (Columns) against **AVG(Profit)** (Rows), you can visually identify the "tipping point" where discounts start to make products unprofitable.

Adding a **Trend Line** from the Analytics pane will make this relationship even clearer.



Case Study 2: Dashboard

Putting It Together

Assemble your three worksheets (Map, Scatter, Bar Chart) onto one dashboard.

Add a **Filter Action** so that clicking on a **Country** on the map filters both the scatter plot and the bar chart.

Insights

Your dashboard will allow the VP of Sales to see *which* countries are unprofitable (from the map) and *why* (by seeing the discount/product mix for that country in the other charts).



Quiz: Case Studies

Q3 Quiz Time

Question

In the HR Attrition dashboard, you want to click on the "Sales" department and see **only** the salary band data for Sales. What kind of Dashboard feature do you need to create?

A) A Parameter, **B) A `FIXED` LOD**, C) A Filter Action, D) A Set

Quiz: Solution

A: C
Solution

Answer: C) A Filter Action

A **Filter Action** allows a worksheet (the "Source Sheet") to act as a filter for other worksheets (the "Target Sheets") on the same dashboard.

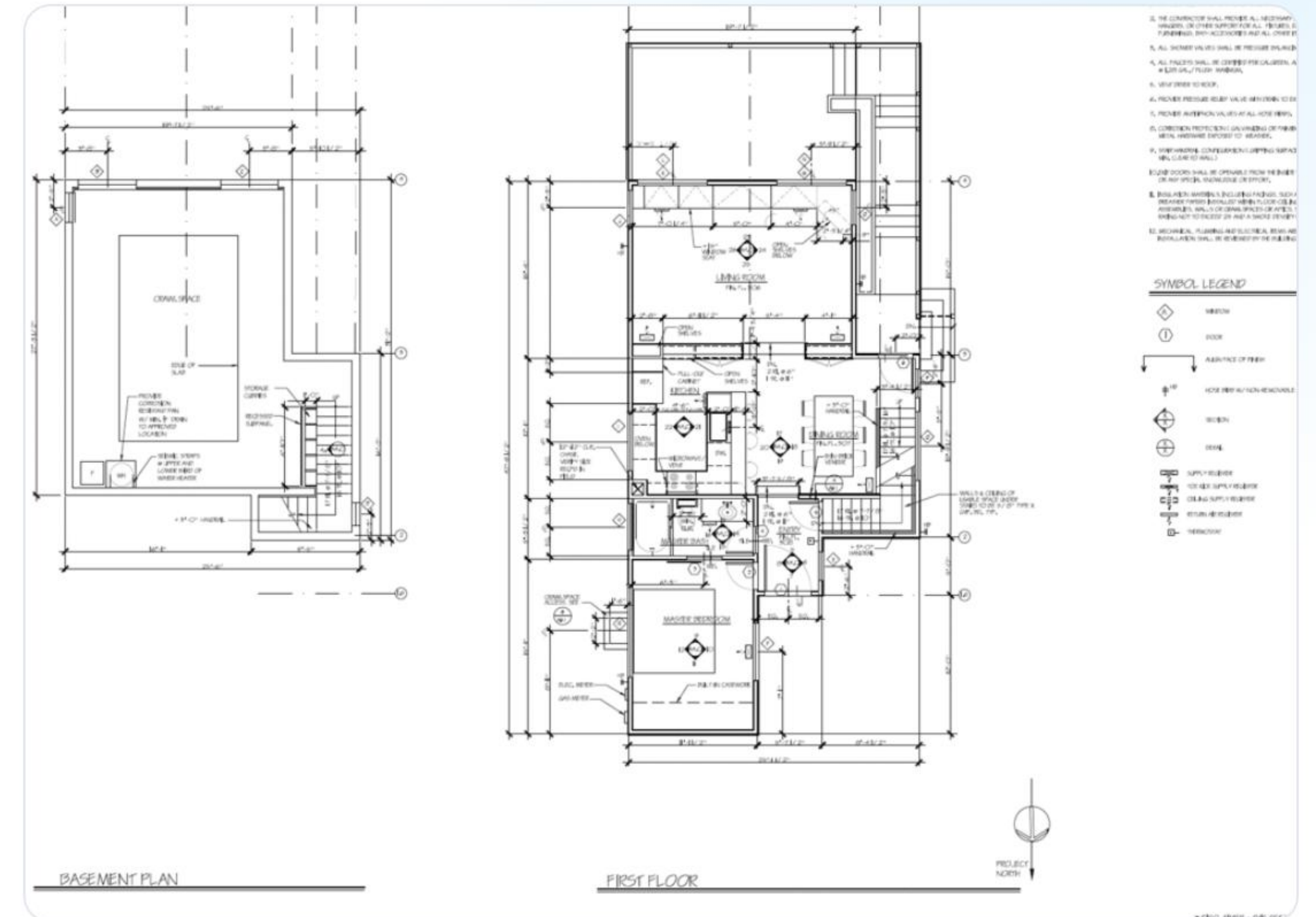
Final Project: Executive Dashboard

Your Capstone Assignment

Combine all skills from Module 1 and 2. Create a single-page executive dashboard for the CEO using **both datasets** (on separate, un-blended charts).

Requirements:

1. **HR KPIs:** Total Employees, Overall Attrition Rate.
2. **Sales KPIs:** Total Sales, Overall Profit Margin.
3. **Viz 1:** Sales & Profit Trend (Dual-Axis Line).
4. **Viz 2:** Attrition by Department (Bar).
5. **Viz 3:** Profit Map by Country.
6. Use a "Top N" Parameter to filter Sales.



Project Best Practices

Formatting is Key

- Use a consistent color palette.
- Format all tooltips to be clean and readable.
- Use descriptive titles for every chart and the dashboard.
- Use layout containers to organize your dashboard cleanly (e.g., a top container for KPIs, a side container for filters).

Tell a Story

Your dashboard should tell a story. What is the key insight? Add a "Text" object to your dashboard with a 1-2 sentence summary of your findings (e.g., "Sales are up, but high attrition in the Sales team is a major risk.").

How to Share Your Work

“

As we are using Tableau Public, the only way to save your work is to publish it to your online profile.

”

*Go to File > Save to Tableau Public As...
(This will make your visualization visible to anyone on the web)*

Module 2 Complete!

You've Leveled Up!

You now have the skills to move beyond simple charts and build true analytical applications. You can control data granularity, build interactive dashboards, and answer complex business questions.

Next Steps:

Get ready for the **Advanced Module**, where we'll cover complex table calculations, forecasting, and advanced analytics features.



Questions?

Q & A

Use this time to ask about LODs, Parameters, or any of the case studies.

Reference: Key Functions

Key Functions

- IF/THEN/ELSE/END
- CASE/WHEN/ELSE/END
- {FIXED [dim] : AGG(measure)}
- {INCLUDE [dim] : AGG(measure)}
- {EXCLUDE [dim] : AGG(measure)}

Key Concepts

- Parameters (User Input)
- Sets (IN/OUT)
- Dual-Axis (Different Scales)
- Context Filters (Pre-LOD)
- Dashboard Actions (Filter/Highlight)

Additional Resources

Keep Learning

The best way to learn Tableau is to practice. Download interesting datasets and try to build dashboards that answer a specific question.

- Tableau Public Gallery
- Makeover Monday
- Workout Wednesday

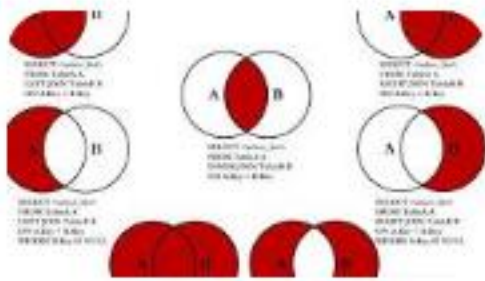


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Image Sources



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